

ENTERPRISE SYSTEM PROPOSAL

Heavy Equipment Rental & Fleet Intelligence Platform



One platform for online rentals, real-time machine tracking, and bank-grade financial control — built for scale across Ghana and West Africa.

THE OPPORTUNITY

Rental demand is booming. Operations haven't kept up.

15-25%

utilization upside from real-time availability & redeployment of idle assets

1-3 days

current quote-to-booking turnaround — customers book whoever answers first

100%

of fleet value exposed without live tracking, geofencing & tamper alerts

Ghana's construction and mining sectors increasingly rent rather than own. **The operators who win are the ones who can prove availability, mobilize fast, and protect their fleet with data.**

CURRENT STATE

Six operational gaps draining revenue today



Manual booking & scheduling

Double-bookings, slow quotes, enquiries lost after hours



No real-time asset visibility

Unauthorized use, site disputes, theft exposure



Estimated engine hours

Revenue leakage and billing disputes on hourly rentals



Reactive maintenance

Unplanned downtime and emergency repair premiums



Fragmented payments

Cash-handling risk and slow, manual reconciliation



No consolidated reporting

Decisions made on incomplete or stale data

THE SOLUTION

One platform. Four integrated pillars.



Rental Commerce

01

Customer portal & apps: catalog, live availability, online booking, digital contracts, MTN MoMo & card payments



Fleet Telematics

02

Live GPS map, geofencing, verified engine hours, route playback, theft prevention & maintenance triggers



Financial Operations

03

Automated invoicing, deposits, maker-checker payouts, ledger and same-day reconciliation



Analytics & Intelligence

04

Executive KPIs, utilization & revenue analytics, scheduled reporting and BI integration

PILLAR 01 — RENTAL COMMERCE

A booking experience customers choose first



Equipment catalog

Full specs, imagery, rate cards and live availability by branch and date



Instant online booking

Guided reservation with conflict checks, haulage & operator add-ons



KYC & digital contracts

Ghana Card / business verification and e-signed rental agreements



Local payments built-in

MTN MoMo, Telecel Cash, cards & bank transfer via Paystack — deposits and refunds included



Automated notifications

SMS, email & push for confirmations, receipts, delivery and returns

WHY IT MATTERS

**Every enquiry captured
— 24/7, on any device.**

Corporate accounts get multi-user access, credit terms and consolidated billing — winning larger contractors and repeat business.

***Target: 60%+ of bookings online
within 12 months***

OPERATIONS & ADMINISTRATION

A single command center for the whole operation



Command dashboard

Fleet status, active rentals, pipeline, revenue and alerts in one view



Asset lifecycle

Acquisition to disposal, insurance, documents and per-asset profitability



Booking & dispatch

Approvals, agreements, extensions, delivery and pickup logistics



Maintenance

Engine-hour driven preventive schedules, work orders, service history



Pricing & promotions

Dynamic and seasonal rates, campaigns, governed discount approvals



Governance & RBAC

Fine-grained roles, maker-checker approvals, immutable audit trail

Every machine visible. Every hour verified.



Live fleet map

Real-time position, speed, ignition state and utilization for every asset



Geofencing & alerts

Site boundaries with entry/exit alerts and after-hours movement detection



Verified engine hours

Telemetry-captured hours drive billing, utilization and maintenance — no more estimates



Theft prevention

Tamper & disconnect alerts, recovery-mode tracking, immobilization support



Hardware-agnostic

Teltonika-class devices or Samsara/Geotab APIs — no vendor lock-in

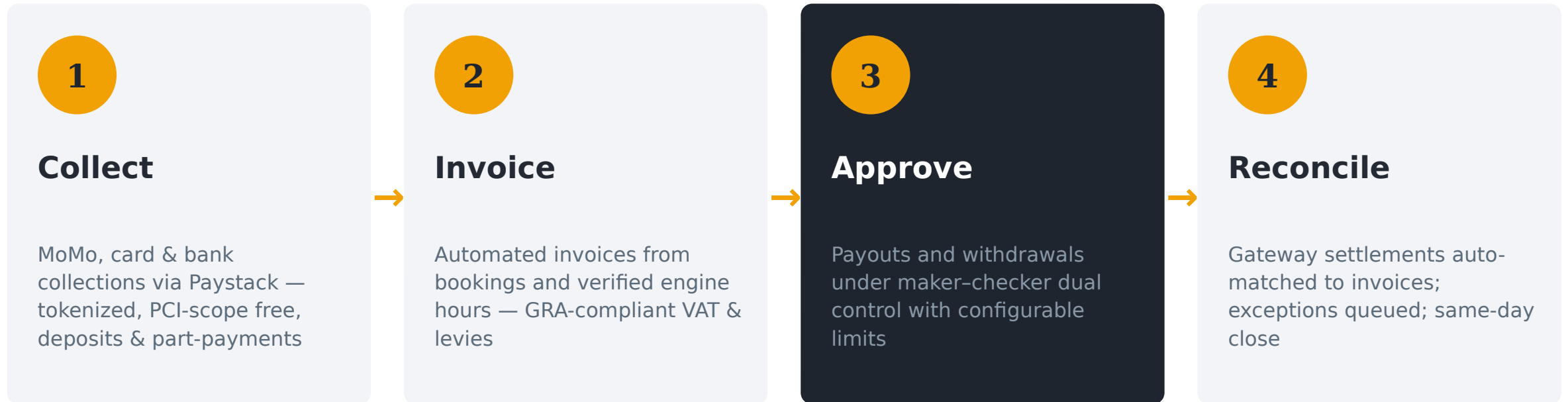


Predictive triggers

Service alerts fire automatically from engine-hour thresholds

Result: billing disputes below 2% of rentals, and 100% of fleet geofenced and alarmed.

Bank-grade control over every cedi



Double-entry ledger · immutable audit trail · exports to QuickBooks, Sage & Xero

Reconciliation time: days of manual work → same-day, automated.

Cloud-native, API-first, built to scale

Experience

Customer portal (Next.js PWA) · Operations console · iOS & Android apps · Field app

API & Edge

API Gateway · WAF & DDoS protection · OAuth 2.0 / OIDC · CDN

Domain Services

Rental & Booking · Fleet Telematics · Billing & Payments · Customer & KYC · Maintenance · Notifications

Data & Events

PostgreSQL + PostGIS · Time-series telemetry store · Redis · Object storage · Event streaming

Integrations

Paystack / MTN MoMo · Samsara / Geotab / Teltonika · SMS & email · Accounting export · Maps

99.9%

availability SLA

≤ 15 min

recovery point (RPO)

≤ 4 hrs

recovery time (RTO)

500+

assets, 100k+ bookings/mo capacity

Engineered for trust from day one



Encryption everywhere

TLS 1.2+ in transit, AES-256 at rest for data, backups and documents



Strong identity

SSO with OAuth 2.0/OIDC, mandatory MFA for admin & finance roles



Hardened perimeter

WAF, DDoS protection, CI vulnerability scanning, annual penetration tests



Full auditability

Immutable logs of every sensitive action — who, what, when, from where

REGULATORY ALIGNMENT



Ghana Data Protection Act, 2012 (Act 843)

Consent capture, subject rights workflows, DPC registration



PCI DSS

Out of scope by design — tokenized payments via certified gateway



Ghana Revenue Authority

VAT/NHIL/GETFund computation, e-VAT-ready invoices

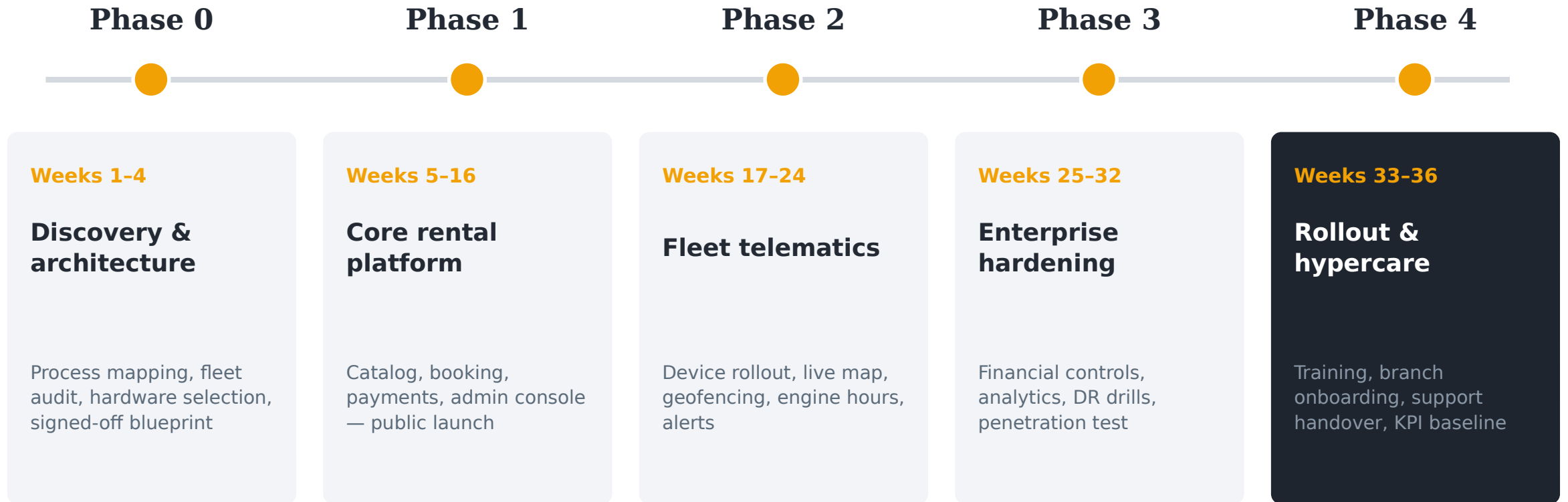


ISO/IEC 27001 practices

Control framework guiding security operations

DELIVERY PLAN

Live in 4 months. Enterprise-grade in 9.



Agile two-week sprints · monthly steering committee · quality gates at every phase

MEASURABLE OUTCOMES

What success looks like at 12 months

$\geq 70\%$

tracked fleet utilization

vs. untracked today

$\geq 60\%$

of bookings made online

vs. 0% today

< 2 hrs

quote-to-booking time

vs. 1-3 days today

$< 2\%$

billing disputes

engine hours verified by telemetry

-25%

unplanned downtime

telemetry-driven maintenance

99.9%

platform availability

monthly SLA

INVESTMENT

Structured investment, refined in discovery

Phase 0 — Discovery & architecture	One-time	\$8k - 15k
Phase 1 — Core rental platform	One-time	\$45k - 70k
Phase 2 — Fleet telematics software	One-time	\$25k - 40k
Phase 3 — Hardening & analytics	One-time	\$20k - 35k
Telematics hardware & install	Per asset	\$150 - 400
Cloud & third-party services	Monthly	\$800 - 2,500
Support & maintenance (SLA)	Monthly	12-18% of build / yr

HOW TO READ THIS

Indicative planning ranges. Fixed commercial terms are issued after the 4-week Phase 0 discovery, once fleet size, branches and hardware are confirmed.

Gateway fees (~1.95% local) are pass-through and can be absorbed or surcharged per pricing policy.

Governed delivery, managed risk

How we govern



Joint steering committee — executive sponsor, operations, finance, delivery — monthly



Two-week sprints with stakeholder demos at every review



Quality gates per phase: automated tests, UAT sign-off, go-live checklists



Structured change management: role-based training and train-the-trainer for branches

Top risks & mitigations



Remote-site connectivity

Store-and-forward devices, multi-network SIMs



Mobile money API downtime

Dual payment rails + offline recording



User adoption

Early involvement, phased rollout, exec sponsorship



Scope creep

Fixed phase gates and change-control board

NEXT STEPS

From proposal to platform in three moves

1

Approve

Executive sign-off on this proposal and its phased scope

2

Discover

4-week Phase 0: requirements, fleet audit, hardware selection, fixed pricing

3

Mobilize

Delivery team starts Phase 1 build immediately after blueprint sign-off

Every machine visible. Every cedi accounted for. Every booking captured.